

Logic Circuits · AS · Answers

24 written + 6 pen-first = 30 · Student copy

One mark per point unless grouped; accept equivalent clear wording. Do not credit contradictory answers. Drawing rubrics require human checking; authored table groupings are stated explicitly.

1. Warning horn [2]

$H = ((\text{NOT } G) \text{ AND } L) \text{ OR } T$. NOT on G only [1]; AND with L, then OR with T [1]. Accept equivalent bracketing. G=0 means not locked.

2. Draw the expression [3]

XOR(A,B) [1]; OR(B,C) [1]; NAND joining the two outputs [1]. Each point requires correct inputs. Preserve supplied structure. Clear repeated labels for the same signal are acceptable; no blanket penalty.

3. Read the circuit [2]

$X = (A \text{ XOR } B) \text{ NOR } (\text{NOT } C)$. Both incoming branches correct [1]; final NOR combining them [1]. Preserve bubble scope.

4. Trace [4]

A NAND B: 11111100; A OR C: 01011111; X: **01011100**. Our scheme: one mark per complete consecutive output pair, 01 / 01 / 11 / 00. Working columns carry no separate marks.

5. Table to expression [2]

$X = ((\text{NOT } A) \text{ AND } (\text{NOT } B)) \text{ OR } (A \text{ AND } B)$. Correct terms for 00 and 11 [1]; OR joining them [1]. Also accept NOT(A XOR B), or any verified equivalent expression using the six specified gates, for 2. Bare XNOR names a function but does not meet the requested gate set.

6. Wiring [2]

Bubble negates the output [1]. A crossing without a dot is not connected in these diagrams [1].

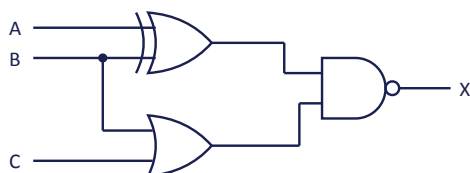
7. Authored exam-shape rehearsal [4]

(a) OR(P,Q) and NAND(R,S), both with correct inputs [1]; NOR joining their outputs [1]. (b) **01110110**: first four 0111 [1], last four 0110 [1]. Our scheme follows the complete-half style: no separate intermediate-column marks.

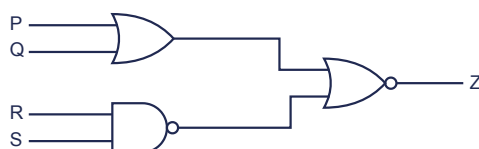
8. Replacement [5]

(a) U: **0110**; V: **0111**. Our scheme: one mark per complete consecutive pair in each column [4]. (b) **A=1, B=1** [1]; U=0, V=1 confirms the difference.

Q2 model



Q7(a) model



9. Pen-first Forge [6]

(a) OR(A,B) feeding AND with C [1]; correct connected, labelled sketch [1]. (b) At 001, intermediate 0, X=0 [1]; at 110, intermediate 1, X=0 [1]. Each point needs both values. (c) Any one of **001 or 110** [1]; target 0 and proposed circuit 1, both [1]. Equivalent clear working accepted.

Unmarked extension: 1+1 = binary 10: Sum 0, Carry 1. One output bit cannot represent the three possible totals 0, 1, 2. A truth table states stable Boolean behaviour, not propagation delay; physical gates/wiring take time.